

Real Time Traffic Simulation

User Guide

Introduction

The Real-Time Traffic Simulation System is a desktop application that visualizes and controls an urban traffic simulation using **SUMO** as the backend and **JavaFX** as the graphical user interface.

The application allows users to:

- View a live traffic simulation on a fixed road network
- Control traffic lights manually
- Enable rush hour traffic
- Spawn emergency vehicles
- Filter and inspect vehicles
- Monitor traffic statistics
- Export simulation data for further analysis



This guide explains how to run and use the application.

System Requirements

Software Requirements

- Java JDK 17 or higher
- SUMO (Simulation of Urban Mobility)
- Operating System:
 - ✓ Windows

Required Files:

The following files must exist in the project resources:

- final_map.net.xml
- final_map.sumocfg

1. Application Structure and Startup

The application consists of two main startup classes:

MainApp

MainApp contains the standard Java `main()` method and serves as the entry point when launching the application.

```
package trafficsimulation;

import trafficsimulation.gui.TrafficSimulationAppFX;
import javafx.application.Application;

public class MainApp { @Mojtaba
    public static void main(String[] args) { Application.launch(TrafficSimulationAppFX.class, args); }
}
```

It initializes the JavaFX runtime and delegates execution to the GUI application.

TrafficSimulationAppFX

TrafficSimulationAppFX is the main JavaFX application class.

It performs the following tasks:

- ❖ Loads the road network from the SUMO .net.xml file
- ❖ Initializes the simulation backend
- ❖ Builds the graphical user interface
- ❖ Connects the map view and control panel
- ❖ Handles application shutdown and exports simulation data

Users normally start the application via MainApp, while all GUI logic is handled internally by TrafficSimulationAppFX.

Once started, the main window opens automatically and loads:

- The road network
- Traffic light positions
- The control panel

When the application window is closed, the following files are automatically exported:

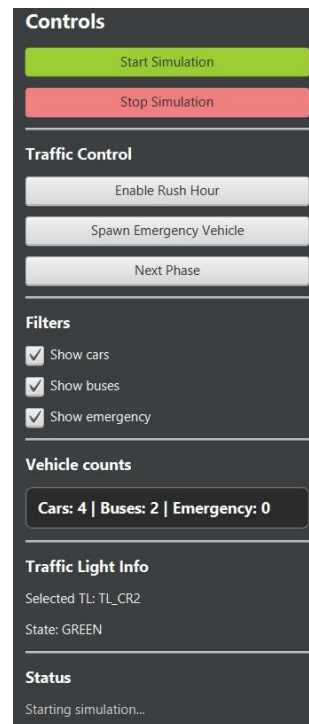
- simulation_stats.csv
- simulation_stats.pdf
- vehicle_positions.csv

2. User Interface Overview

The application window consists of two main areas:

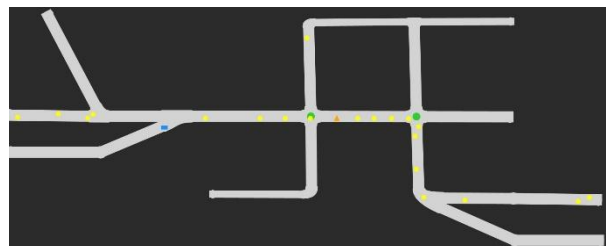
2.2 Control Panel (Right Side)

- Simulation control buttons
- Vehicle filtering options
- Vehicle counters
- Traffic light information panel
- Status



2.1 Map View (Center)

- Displays the road network
- Shows junctions and traffic lights
- Renders vehicles in real time
- Supports zooming and panning



3. Map Interaction

Zoom and Pan

- **Mouse Wheel:** Zoom in / out
- **Right Mouse Button + Drag:** Pan the map
- **Double Click:** Reset zoom and pan

Selection

- **Click a vehicle:** Select and highlight it with a red border
- **Click a traffic light:** Select it for inspection or control
- **Click empty space:** Clear selection

4. Vehicle Visualization

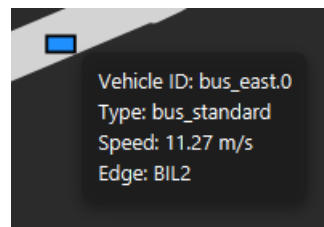
Vehicles are rendered using different shapes and colors:

Vehicle Type	Shape	Color
Car	Circle	Yellow
Bus	Rectangle	Blue
Emergency Vehicle	Triangle	Orange

Vehicle Tooltips

Hovering over a vehicle displays:

- Vehicle ID
- Vehicle type
- Current speed
- Current road segment (edge)

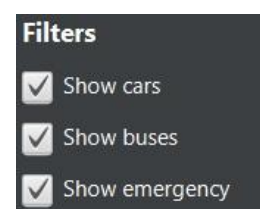


5. Vehicle Filtering and Grouping

Filtering

The control panel provides vehicle filtering options:

- Show cars
- Show buses
- Show emergency vehicles



Users can enable or disable each category independently.

Vehicle Counters

A live counter shows:

Cars

Buses

Emergency

These values update in real time.



6. Simulation Controls

Start Simulation

- Starts the SUMO backend
- Begins live rendering
- Starts statistics collection

Stop Simulation

- Stops the simulation
- Closes the SUMO connection
- Freezes the current state



7. Traffic Control Features

Enable Rush Hour

- Increases vehicle traffic density
- Simulates peak traffic conditions

Spawn Emergency Vehicle

- Adds an emergency vehicle to the simulation
- Emergency vehicles are visually distinct

Manual Traffic Light Control

1. Select a traffic light on the map
2. Click Next Phase
3. The traffic light switches to the next phase



8. Traffic Light Information Panel

Displays information for the selected traffic light:

- Traffic light ID
- Current simplified state:
 - GREEN
 - YELLOW
 - RED



9. Statistics and Data Collection

Automatic Statistics Collection

Statistics are collected automatically during simulation runtime, including:

- Number of vehicles
- Average speed
- Minimum and maximum speed
- Vehicle density per road segment
- Vehicle positions per simulation step

No user interaction is required to start data collection.

10. Exported Files

When the application window is closed, the following files are generated automatically:

File	Description
simulation_stats.csv	Per-step global statistics
simulation_stats.pdf	PDF summary report
vehicle_positions.csv	Vehicle positions and speeds per step

These files can be used for further analysis or reporting.

11. Application Shutdown

When closing the application:

- The simulation is stopped safely
- All collected statistics are exported automatically
- No manual save action is required

12. Error Handling

- ✓ Buttons are disabled or show messages if the simulation is not running
- ✓ If SUMO closes unexpectedly, the simulation stops automatically
- ✓ Status messages appear at the bottom of the control panel

Status

Start the simulation first.

13. Known Limitations

- The road network is fixed and cannot be edited inside the application
- SUMO must be installed correctly for the simulation to run
- Manual traffic light control affects entire junction logic, not individual lanes

15. Conclusion

The Real-Time Traffic Simulation System provides an interactive environment to visualize, control, and analyze urban traffic behavior.

It combines real-time simulation, intuitive controls, and automatic data export to support both experimentation and analysis.